Franka Panda Control Using MediaPipe Body Pose and Hand Gestures

ROS 2 Humble, MoveIt Servo and Franka Hand

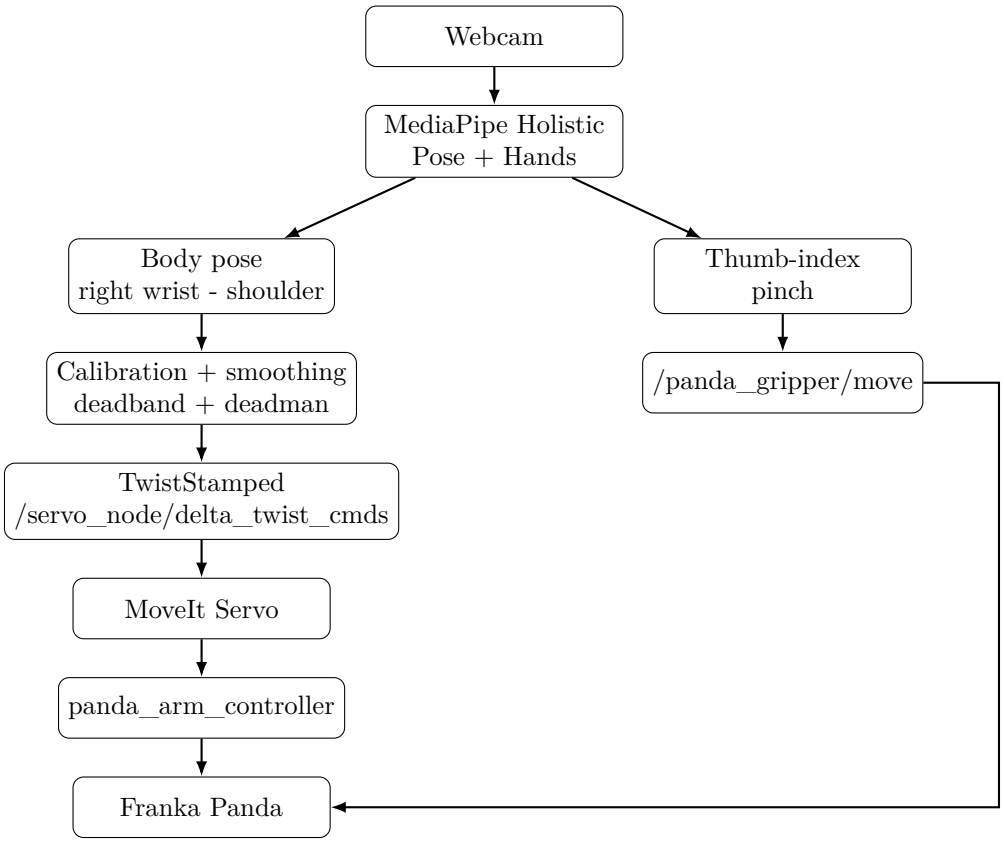
Laboratory Project Guide

16 September 2026

1 Purpose

This project extends the validated Franka Panda ROS 2/MoveIt setup with a camera-based human–robot interface. MediaPipe tracks the operator's body and hands. Right-arm body motion generates small Cartesian velocity commands for MoveIt Servo, while a thumb–index pinch controls the Franka gripper. All motion starts disabled.

2 Architecture



MoveIt Servo receives fresh timestamped Cartesian commands and is configured to output `JointTrajectory` messages to `/panda_arm_controller/joint_trajectory`, with collision checking and singularity/joint-limit handling enabled.

3 Gesture mapping and safety

Input	Result
Right hand toward/away camera	Panda X translation
Right hand left/right	Panda Y translation
Right hand up/down	Panda Z translation
Left wrist above shoulder	Arm deadman enabled
Left wrist lowered	Zero arm velocity
Thumb-index pinch	Gripper closes to 0.04 m
Thumb-index apart	Gripper opens to 0.08 m

The arm additionally requires `C` calibration and keyboard `A`. Default maximum Cartesian speed is 0.03 m/s. Angular commands are disabled. Gripper gestures are separately enabled with `G`.

4 Installation

```
cd ~
unzip Franka_MediaPipe_BodyPose_HandGesture_Control.zip
cd Franka_MediaPipe_BodyPose_HandGesture_Control
conda deactivate
bash scripts/install.sh
```

The installer creates a MediaPipe virtual environment and builds `/panda_body_servo_ws`. It does not modify apt packages automatically.

5 Terminal procedure

In Franka Desk recover/unlock, activate FCI, clear the workspace and verify `ping -c 3 172.16.0.2`.

Terminal 1: Panda + MoveIt + gripper

```
bash scripts/start_robot.sh
```

Terminal 2: MoveIt Servo

```
bash scripts/start_servo.sh
```

Terminal 3: check and enable Servo

```
bash scripts/check_system.sh
bash scripts/enable_servo.sh
```

Expected controller state is `joint_state_broadcaster active` and `panda_arm_controller active`. The earlier Panda test produced approximately 30 Hz joint states and successful gripper actions. The new Servo layer must be validated before normal use.

Terminal 4: MediaPipe

```
bash scripts/run_dry_demo.sh
# after dry validation:
bash scripts/run_real_control.sh
```

6 First real test

Press `C` while standing neutrally. Test the gripper first with `G`. Next press `A`; the arm still cannot move until the left wrist is above the left shoulder. Raise the left hand, move the right hand only a few centimetres, then lower the left hand to stop. If an axis is reversed, change its sign in `config/body_control.json` only after returning to dry mode.

7 Key configuration

```
"max_linear_speed_mps": 0.03,
"body_deadband_m": 0.04,
"body_full_scale_m": 0.22,
"robot_x_from_body_z_sign": -1.0,
"robot_y_from_body_x_sign": 1.0,
"robot_z_from_body_y_sign": -1.0
```

8 Stop

```
# Q in camera window
bash scripts/stop_servo.sh
```

Then stop Servo and MoveIt with `Ctrl+C` and return the Panda to a safe state in Franka Desk.

9 Validation status

ROS 2 hardware communication, MoveIt startup, the active Panda trajectory controller, joint-state streaming and gripper actions were already validated on the lab Panda. The body-pose-to-Servo layer is the next experimental integration and should be validated progressively: dry camera test, Servo checks, zero/small commands, then normal teleoperation.